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Association between the big five personalities and academic performance among grade 12 students at international high school in Thailand

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The well-known Big Five personalities represent five personality characteristics including conscientiousness, extraversion, agreeableness, neuroticism, and openness. Previous literature confirmed that academic performance is associated with conscientiousness personality but there were some uncertainties in other personalities across study populations. The questionnaire-based study was conducted among Grade-12 students at an international high school in Thailand. The data on the Big Five Inventory (BFI) personality assessment, and other baseline characteristics were collected via online or paper questionnaires. One-hundred and thirty-three respondents were female with a median GPA of 3.61. There were 102 respondents with a GPA ≥ 3.61 in the academic achievement (AA)-group and 101 respondents with a GPA < 3.61 in the academic non-achievement (AN)-group. The AA-group had a significantly higher score in the conscientiousness domain while the score in the other four domains was comparable between both groups. Multivariate analysis identified the Conscientiousness personality domain and the female gender as the positively associated factors for academic achievement. Furthermore, a higher BMI and having extensive physical activities were identified as the negatively associated factors for academic achievement. Our study confirmed the universal association between conscientiousness personality and academic achievement among international students in Thailand. The study results would be helpful for policymaker to improve the educational program.

Background

Over the past decades, many personality theories have been conceptualized by expert psychologists. In 1981, Goldberg chose the title "Big Five" to describe five major personality domains. The title was chosen to represent five specific and distinct personality characteristics including conscientiousness, extraversion, agreeableness, neuroticism (the opposite pole of emotional stability), and openness[1] Conscientiousness is a personality trait that focuses on the ability to be thoughtful and well-planned. Extraversion is a personality trait that focuses on sociability and outgoingness. Agreeableness is a personality trait that focuses on trust and kindness towards others. Neuroticism is a personality trait that focuses on emotional instability to negative stimuli. Openness is a personality trait that focuses on creativity and imagination.

The Big Five Inventory (BFI), a well-established personality assessment tool[1], is a 44-item questionnaire that assesses the five personalities attributed to the Big Five Personality. A questionnaire respondent would have to rate their agreement with each of the 44 statements on a scale from "strongly disagree, 0 points" to "strongly agree, 5 points". The scores are subsequently calculated to determine the mean for each of the 5 personality traits. Previous studies have demonstrated its reliability and validity, establishing its strong psychometric properties[2–4].

Previous literature confirmed that academic performance is significantly associated with some specific personality domains[5–14] Another study of university students from various study fields (health science, scientific science, social science, and arts and humanities) was conducted to evaluate their Big Five personalities,

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Intelligence Quotient (IQ) and cognitive competencies (by using the literacy test and the numeracy test)[10] The study confirmed that IQ was strongly associated with numeracy performance while personality was strongly associated with literacy performance. Additionally, the effects of personality and IQ on cognitive competencies were modified by the level of education (freshmen, undergraduates, and post-graduates). The association was stronger in freshmen or undergraduates when compared to post-graduates.

A recent meta-analysis of more than 400,000 students confirmed that the conscientiousness personality was the most robust predictor for the academic performance of students at all school levels[7] In this study, openness, extraversion, and agreeableness personalities were considered weak predictors for academic performance. The strength of association between openness personality and academic performance was stronger at the elementary school level when compared to higher education. This meta-analysis did not find any association between the neuroticism personality and academic performance.

Of all Big Five personality traits, the conscientiousness personality is universally considered the strongest predictor for good academic performance at all school levels[6–11,13–15] However, the association between other Big Five personality traits and academic performance is significant in some study populations[6,7,9,10,15] Based on these findings, some uncertainties in the impact of personality traits on academic performance still exist. Also, there are some effect modifiers such as educational level (elementary school, secondary school, high school, undergraduate, post-graduate level) or field of study (art, science, medicine). Furthermore, most of the previous studies were conducted in developed countries (i.e. United States, and European countries) and the standard school. Until now, there has never been any study focused on students in an international school or those who reside in a developing country.

Given these considerations, we hypothesize an association between academic achievement and all Big Five personality domains, especially Conscientiousness. We also hypothesize that academic achievement will be associated with intrinsic factors such as family background, physical activity, and extracurricular activities.

Methods

Study design and setting

From 1st – 30th September 2023, the questionnaire-based study was conducted among Grade-12 students at an international high school located in Nakhon Pathom province, Thailand. This school teaches three grades; grade 10, grade 11, and grade 12. There were 12 classes per grade with approximately 20 students per class. The name of the school remained anonymous per request of the school administrative board.

The primary objective of this study was to determine the association between the student's academic performances and their personalities. The secondary objective was to determine other baseline characteristics and extrinsic factors associated with the academic performance. The research protocol was approved by the Mahidol University Multi-faculty Cooperative institutional review board with a waiver of informed consent (MU-MOU CoA No. 564/2023). This waiver was granted because completing the anonymous questionnaire is considered as providing voluntary consent to participate in the study.

Questionnaire development

The questionnaire consisted of four sections: (1) the Big Five Inventory (BFI) personality assessment, (2) baseline characteristics (age, gender, race, weight, height, eye-sight, socioeconomic status, and parents' profession and educational level), (3) daily activities (duration of sleep, duration of self-study, duration of physical engagement, duration of social media engagement and a number of meals per day), and (4) academic-related parameters (the latest grade point average or GPA, fields of academic interest and dream profession). All questions were in English and contained both closed-ended and open-ended questions. The draft questionnaire was evaluated by a consultant psychiatrist and a group of international high school students. The questionnaire was subsequently revised for simplicity and understandability. The final version of the questionnaire is available in the supplementary file (online).

Questionnaire distribution

The questionnaire was distributed to all Grade-12 students by their class teacher. An invitation letter with a QR code embed a link to an online questionnaire and a paper questionnaire were simultaneously provided. Students can choose to respond via either the online questionnaire or the paper questionnaire. Participation was voluntary and anonymous. The respondents could skip any questions if they feel uncomfortable to answer. After completing the questionnaire, all respondents were asked to enter their email addresses to take part in a lucky draw, in which there were 5 prizes worth 1,000 Thai baht each (or approximately 20 British pounds).

Statistical analysis

The data from both online and paper questionnaires was collected and stored in the same electronic database. Categorical variables were reported as frequencies and percentages while continuous variables were reported as mean ± SD or median and range according to the distribution.

The study population was categorized into two groups; the academic achievement group (those with an average GPA ≥ 50th percentile; the AA-group), and the academic non-achievement group (those with an average GPA < 50th percentile; the AN-group), All variables of each group were evaluated and then compared. Depending on the data distribution, the Student-t-test or Man-Whitney U test was used to compare the continuous variables, and the chi-square or Fisher-exact test was used to compare the categorical variables. Multivariable analysis with a forward stepwise approach was subsequently performed to identify the independently associated factors for academic achievement.

All analyses were performed with STATA/IC version 14.0 (STATA Corp, College Station, TX). A p-value of < 0.05 is determined to be statistically significant.

Results

During the study period, there were 250 grade-12 students who received the study questionnaire, and the response rate was 81.2% (203/250). Of these 203 respondents, 134 of them were female with a mean age (± SD) of 17.19 ± 0.53 years. The median GPA [Interquartile range (IQR)] of respondents was 3.61 [3.33–3.80] while the mean GPA (± SD) was 3.53 ± 0.39. By using the median GPA as the cut-off value, 102 respondents with a GPA of 3.61 or more were enrolled into the AA-group, and 101 respondents with a GPA less than 3.61 were enrolled into the AN-group. Table 1 shows the characteristics of respondents in the AA-group and the AN-group.

Baseline characteristics

The mean age (± SD) of the AA-group and the AN-group were comparable (17.15 ± 0.50 years vs. 17.24 ± 0.57 years; p-value = 0.21). The respondents in the AA-group had a significantly higher proportion of females (78.4% vs. 53.5%; p-value < 0.001), a significantly lower mean body weight (53.70 ± 8.33 vs. 59.65 ± 11.08; p-value < 0.001), a significantly lower mean body height (163.63 ± 7.08 vs. 167.84 ± 78; *p-value* < 0.001), and a significantly lower mean BMI (20.02 ± 2.56 vs. 21.15 ± 3.54; p-value = 0.01) when comparing with the AN-group.

Big five personality

Table 2 presents the mean score (±) for each of the five domains in the Big Five personality.

between the two groups. The respondents in the AA-group had a significantly higher score in the conscientiousness domain (3.31 ± 0.56 vs. 3.12 ± 0.58, p-value = 0.02) and the slightly higher score in the agreeableness domain (3.59 ± 0.59 vs. 3.50 ± 0.55, p-value = 0.31) than the AN-group. However, the score in the other three domains including the extraversion domain (3.10 ± 0.68 vs. 3.09 ± 0.66, p-value = 0.93), the neuroticism domain (3.21 ± 0.65 vs. 3.17 ± 0.65, p-value = 0.64), and the openness domain (3.46 ± 0.63 vs. 3.52 ± 0.54, p-value = 0.46) were comparable between two groups.

Family parameters

The majority of respondents' parent marital status was married, which is comparable between the two groups (85.4% vs. 80.1%; p-value = 0.37). The business owner was the most common father's occupation (61.8% vs. 73.3%; p-value = 0.17) and mother's occupation (45.1% vs. 49.5%; p-value = 0.47) in both AA- and AN-groups. The respondents in the AA-group were more likely to have a mother with a bachelor's degree or higher education level (91.2% vs. 81.2%; p-value = 0.04).

Daily activities

The respondents in the AA-group were significantly more likely to spend more than two hours on self-studying (23.0% vs. 11.1%; p-value = 0.03) and significantly less likely to spend more than two hours on physical activities (9.9% vs. 16.2%; p-value = 0.004). The proportion of respondents who had less than three meals per day (31.1% vs. 25.5%; p-value = 0.38), spending of time to sleep more than eight hours per day (3.0% vs. 6.9%; p-value 0.21), spending of time for social medial use more than 2 h per day (66.1% vs. 71.6%; p-value = 0.31), and spending of time for long journey to school more than 30 min per trip (25.2% vs. 28.4%; p-value = 0.21) were comparable between two groups.

Academic-related parameters

The majority of respondents in the AA-group felt satisfied or very satisfied with their GPA (60.8%) while only 26.7% of respondents in the AN-group felt satisfied with their GPA p-value < 0.001). The most chosen extracurricular activities were the Science (56.9% vs. 45.5%; p-value = 0.11), followed by the Social Science (39.2% vs. 42.6%; p-value = 0.63) which were similar between the two groups. The respondents in the AA-group were significantly more likely to choose the doctor career (27.5% vs. 9.9%; p-value < 0.001) and significantly less likely to choose the artist career (16.7% vs. 29.7%; p-value = 0.03) as their dream career.

Independently associated factors for academic achievement

Multivariate analysis (adjusted Odds Ratio [95% confidence interval]; p-value) identified the Conscientiousness personality domain (1.89 [1.12–3.20; *p-value* = .02]) and the female gender (2.66 [1.39–5.09; *p-value* = .003]) as the positively associated factors for academic achievement. Furthermore, a higher BMI (0.90 [0.81–0.99; *p-value* = .03]) and having extensive physical activities (0.30 [0.09–0.99; *p-value* = .05]) were identified as the negatively associated factors for academic achievement. Table 3 shows the results from multivariate analysis.

Discussion

Our study explored the difference in Big Fiver personality between the academic achievement student (a GPA ≥ 50th percentile) and the academic non-achievement student (a GPA < 50th percentile). The study results showed a significant difference in one of the Big Five personality "Conscientiousness" domains. Additionally, univariate analysis determined significant differences in some baseline characteristics (including gender, body weight, height, and BMI) and some family and activity factors (including the level of the mother's education, the self-study duration, the physical engagement duration, and the academic field of interest).

Similar to previous literature that has documented how female students have a higher tendency to be academically achieved compared with male students[7,13] It is therefore conclusive that the higher proportion of female students in the AA-group could explain the lower body weight, the lower height, and the lower BMI

Variables	All (*N* = 203)	AA-group (*n* = 102)	AN-group (*n* = 101)	*p* value
Baseline characteristics				
Mean age (± SD), year	17.19 ± 0.53	17.15 ± 0.50	17.24 ± 0.57	0.21
Female gender	134 (66.0%)	80 (78.4%)	54 (53.5%)	< 0.0010
Mean body weight (± SD), kg	56.66 ± 10.22	53.70 ± 8.33	59.65 ± 11.08	< 0.0001
Mean height (± SD), cm	165.72 ± 7.58	163.63 ± 7.08	167.84 ± 7.50	0.0001
Body mass index (kg/sq.m)	20.58 ± 3.13	20.02 ± 2.56	21.15 ± 3.54	0.01
Nationality				
Southeast Asians	194 (95.6%)	100 (98.0%)	94 (93.1%)	0.17*
East Asians	8 (3.9%)	2 (2.0%)	6 (5.9%)	
Others	1 (0.5%)	0	1 (1.0%)	
Having short-sightedness	134 (66.0%)	62 (61.4%)	72 (70.6%)	0.17
Family parameters				
Mean number of siblings (± SD)	1.11 ± 0.88	1.07 ± 0.81	1.14 ± 0.94	0.55
Marital status				
Married	169 (83.3%)	88 (85.4%)	81 (80.1%)	0.37*
Divorced or separated	22 (10.8%)	9 (8.7%)	13 (12.9%)	
Widowed	2 (1.0%)	2 (2.0%)	0	
Never married	6 (3.0%)	2 (2.0%)	4 (4.0%)	
Missing	4 (2.0%)	1 (1.0%)	3 (3.0%)	
Father occupation				
Doctor	7 (3.5%)	5 (4.9%)	2 (2.0%)	0.17*
Engineer	11 (5.4%)	8 (7.8%)	3 (3.0%)	
Business owner	137 (67.5%)	63 (61.8%)	74 (73.3%)	
Unemployed or housewife	12 (5.9%)	8 (7.8%)	4 (4.0%)	
Others	36 (17.7%)	18 (17.7%)	18 (17.8%)	
Mother occupation				
Doctor	4 (2.0%)	2 (2.0%)	2 (2.0%)	0.47*
Engineer	2 (1.0%)	2 (2.0%)	0	
Business owner	96 (47.3%)	46 (45.1%)	50 (49.5%)	
Unemployed or housewife	61 (30.0%)	28 (27.5%)	33 (32.7%)	
Others	40 (19.7%)	24 (23.6%)	16 (15.8%)	
Father with Bachelor's degree or higher	166 81.8%)	87 (85.3%)	79 (78.2%)	0.19
Mother with Bachelor's degree or higher	175 (86.2%)	93 (91.2%)	82 (81.2%)	0.04
Daily activities				
Less than three meals per day	58 (28.6%)	32 (31.1%)	26 (25.5%)	0.38
Longer sleeper (8 h or more per day)	10 (4.9%)	3 (3.0%)	7 (6.9%)	0.21*
Extensive self-study (2 h or more per day)	34 (17.1%)	23 (23.0%)	11 (11.1%)	0.03
Extensive physical activities (2 h or more per day)	20 (10.0%)	10 (9.9%)	16 (16.2%)	0.004
Extensive social media use (2 h or more per day)	141 (69.5%)	68 (66.1%)	73 (71.6%)	0.31
Long journey to school (30 min or more per trip)	55 (27.1%)	26 (25.2%)	29 (28.4%)	0.21
Academic-related parameters				
Mean grade point average (± SD)	3.53 ± 0.39	3.82 ± 0.11	3.24 ± 0.35	< 0.001
Median grade point average [range]	3.61 [2.00–4.00]	3.80 [3.61-4.00]	3.33 [2.00-3.60]	< 0.001*
Satisfaction in their grade point average				
Very unsatisfied	11 (5.5%)	5 (4.9%)	6 (5.9%)	< 0.001*
Unsatisfied	26 (13.1%)		17 (16.7%)	
Neutral	73 (36.7%)	26 (25.5%)	47 (46.1%)	
Satisfied	85 (42.7%)	58 (56.9%)	27 (26.5%)	
Very satisfied	4 (2.0%)	4 (3.9%)	0	
Missing	6 (2.9%)	1 (1.0%)	5 (4.9%)	
Chosen extra-curricular activities				
Science	104 (51.2%)	58 (56.9%)	46 (45.5%)	0.11
Social science	83 (40.9%)	40 (39.2%)	43 (42.6%)	0.63
Language	33 (16.3%)	19 (18.6%)	14 (13.9%)	0.36
Informative technology	29 (14.3%)	15 (14.7%)	14 (13.9%)	0.86
Continued				

Variables	All ($N = 203$)	AA-group ($n = 102$)	AN-group ($n = 101$)	p value
Art	20 (9.9%)	10 (9.8%)	10 (9.9%)	0.98
Dream occupation				
Business career	101 (49.8%)	46 (45.1%)	55 (54.5%)	0.18
Artist career	47 (23.2%)	17 (16.7%)	30 (29.7%)	0.03
Doctor career	38 (18.7%)	28 (27.5%)	10 (9.9%)	0.001
Engineer career	36 (17.7%)	23 (22.6%)	13 (12.9%)	0.07
Lawyer career	13 (6.4%)	7 (6.9%)	6 (5.9%)	0.79

Table 1. Characteristics of the respondents in the academic achievement (AA) group and the academic non-achievement (AN) group. * Non-parametric statistics were used.

Mean scores (± SD)	All ($N = 203$)	AN-group ($n = 101$)	AA-group ($n = 102$)	p value
Conscientiousness	3.21 ± 0.58	3.12 ± 0.58	3.31 ± 0.56	0.02
Extraversion	3.09 ± 0.67	3.09 ± 0.66	3.10 ± 0.68	0.93
Agreeableness	3.55 ± 0.57	3.50 ± 0.55	3.59 ± 0.59	0.31
Neuroticism	3.19 ± 0.65	3.17 ± 0.65	3.21 ± 0.65	0.64
Openness	3.49 ± 0.59	3.52 ± 0.54	3.46 ± 0.63	0.46

Table 2. Big five personality scores of the respondents in the academic achievement (AA)-group and the academic non-achievement (AN)-group.

Independent factors	Unadjusted odds ratio [95% CI; p-value]	Adjusted odds ratio [95% CI; p-value]
Conscientiousness	1.79 [1.09–2.94; $p = 0.22$]	1.89 [1.12–3.20; $p = 0.02$]
Female gender	3.16 [1.71–5.84; $p < 0.001$]	2.66 [1.39–5.09; $p = 0003$]
BMI	0.86 [0.81–0.97; $p = 0.01$]	0.90 [0.81–0.99; $p = 0.03$]
Extensive physical activities (2 h or more per day)	0.22 [0.07–0.67; $p = 0.008$]	0.30 [0.09–0.99; $p = 0.05$]

Table 3. Independently associated factors for academic achievement from the multivariate analysis.

in the AA-group. When traits such as height or weight are assessed individually, women tend to be smaller and lighter compared to men. After considering BMI which accounts for the relative proportion of weight and height, the differences become less significant. After performing multivariate analysis, the lower body weight and height were no longer independently associated with the AA-group.

Conscientiousness personality is the only documented personality trait to be significantly associated with academic achievement in our study. Previous studies have suggested that personality traits have less impact on academic performance with increasing age[7] The assumption is that in the university setting, the teachers are less personally involved with the students. Therefore, the teachers have less opportunity to be affected by the students' personalities and characteristics. Furthermore, the association may also depend on the school curriculum and the way they evaluate their students. While the Thai curriculum tends to focus on the memorization of subject matter, the international curriculum focuses more on developing students' ability to think and solve problems critically. These findings could be interpreted in two ways. First, the students who aim to be academically successful should exhibit some levels of discipline and responsibility. On the other hand, the education system tends to reward those who are endorsed with conscientiousness above all other fundamental characteristics (i.e. creativity, cooperation, etc.). Therefore, the school's evaluation system should be revised to also reward those with good characteristics other than conscientiousness.

One interesting finding was the students in the AA-group were significantly associatedwith having a mother who graduated with a bachelor's degree or higher, but no association with the father's education level. This may be explained that in Asian countries including Thailand, a considerable proportion of housewife mothers take care of their children while the father goes out to work. Therefore, a well-educated mother would have more opportunities to positively influence their children far greater than that of a well-educated father. However, the positive association between the mother's educational level and academic achievement disappeared in the multivariate model. Given that the majority of mothers in both groups graduated with a bachelor's degree or higher (> 80%), it would be difficult to confirm the significant association after adjusting for other factors. Therefore, this interesting association should be further explored in other school settings in which the student's parents have heterogeneity in the parent's education level.

When focused on the daily activities, it is not surprising to reveal that those in the AA-group were significantly more likely to have extensive self-study and less likely to have extensive physical activities. Given these two daily

activities may be negatively correlated, only having extensive physical activities was identified as the associated factor in the multivariate model. Furthermore, it is necessary to emphasize that more than two-thirds of students in both groups similarly spent more than 2 h per day in using social media. Based on our study findings, the use of social media should not be stigmatized as a destructive activity against student academic performance as documented in a recent meta-analysis[16].

For the dream career, those in the AA-group were significantly more likely to choose a medical career and less likely to choose an artistry career. This could be explained by the fact that our study used the median GPA to distinguish between the AA-group and the AN-group. Additionally, the GPA is primarily science and mathematics-based, two of the most valuable skills desired in a medical career. Therefore, it is of no surprise that those in the AA-group would be more interested in a medical career and less interested in an artistry career.

Our study has some strengths to be mentioned. First, this was the first study to explore the association between the personality domain and academic achievement among high school students in a developing country. Second, we did not only explore Big Five personality domains but also focused on other baseline characteristics, family parameters, daily activities, and other academic-related parameters. Lastly, the response rate was very high (81.2%) when compared with other questionnaire-based studies conducted among high school students.

Our study has some potential limitations. First, the study was conducted at an international school in Thailand. Therefore, the study results may not be applicable to students in other types of schools (i.e. government schools, schools in other countries). Second, the response bias may be an issue. Given that the respondents had to submit the paper questionnaire to their class teacher, some students may answer some questions positively. Lastly, the median GPA was used to distinguish between the AA-group and the AN-group. This may not be the best way to categorize but it was not the most feasible one.

Conclusions

Our study confirmed the universal association between conscientiousness personality and academic achievement among international students in developing countries such as Thailand. Furthermore, we also identified the positive association between academic achievement and female gender and the negative association with lower BMI and having extensive physical activities. The results of this study would be useful for the school policymaker to comprehensively understand the population, to improve the educational strategies and evaluation systems. This would ultimately result in future success in the student's goals and career.

Data availability

The study dataset is available from the corresponding author upon reasonable request.

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Author contributions
P.R. and G.S. conceptualized and supervised the study. K.B. prepared and distributed the questionnaire. P.R. and K.B. performed analysis and wrote the first draft of manuscript. All authors reviewed and approved the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

Ethics approval and consent to participate
The study was approved Mahidol University Multi-Faculty Cooperative institutional review board with a waiver of informed consent (MU-MOU CoA No. 564/2023). The study has been performed according to the principles of the Declaration of Helsinki.

Consent to publish
Not applicable.

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